Project Title: Network Monitoring and Alert System

Problem Statement:
Build a Network Monitoring and Alert System that monitors network device availability and response time using network sockets and monitoring utilities. Integrate the application with an SDN controller to dynamically monitor network conditions, identify connectivity issues, and apply traffic-management policies when required.

Project Expectations

- Monitor multiple network devices and hosts.
- Check device availability and connectivity.
- Measure response time and detect delays.
- Generate alerts when devices become unreachable.
- Maintain logs of network status and events.
- Establish communication using TCP/UDP sockets wherever applicable.
- Implement a Mininet-based network topology.
- Integrate the monitoring application with an SDN controller.
- Monitor traffic statistics and classify network flows.
- Apply SDN policies based on network conditions.
- Measure monitoring accuracy, response time, and scalability.

Protocol: TCP/UDP (as applicable)
SDN Platform: Mininet + SDN Controller (e.g., Ryu or OpenFlow-compatible controller)

Project Development Flow

1. Problem Understanding & Architecture
2. Network Monitoring and Socket Implementation
3. Mininet + SDN Implementation
4. Monitoring Application–SDN Integration
5. Initial Testing & Demo
6. Complete Application Functionality
7. Dynamic SDN Monitoring and Policy Enforcement
8. Challenging Scenarios / Robustness Testing
9. Performance Evaluation & Comparison

Protocol Responsibilities

Protocol	Responsibility	Example
TCP	Reliable, connection-oriented communication between the monitoring application and monitoring agents/servers.	Send monitoring commands, receive status reports, and transfer logs.
UDP	Lightweight, connectionless communication for periodic status updates or monitoring probes where low overhead is preferred.	Send heartbeat messages and collect simple status updates.
ICMP (Ping)	Check whether a network host is reachable and estimate round-trip time.	Detect whether a router or host responds to ping requests.

Important: Ping uses ICMP, not TCP or UDP. TCP and UDP are transport-layer protocols, while ICMP is used for network-layer control and diagnostics.

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